Code:

```
#include <iostream>
using namespace std;

void select_sort(int [], int);
void swap(int *, int *);

int main()
{
    int a[] = {5, 3, 1, 2, 4};
    int n = 5, i;

    select_sort(a, n);

    for(i = 0; i < n; i++)
        cout << a[i] << " ";

    return 0;
}

void select_sort(int a[], int n)
{
    int i, j, min;

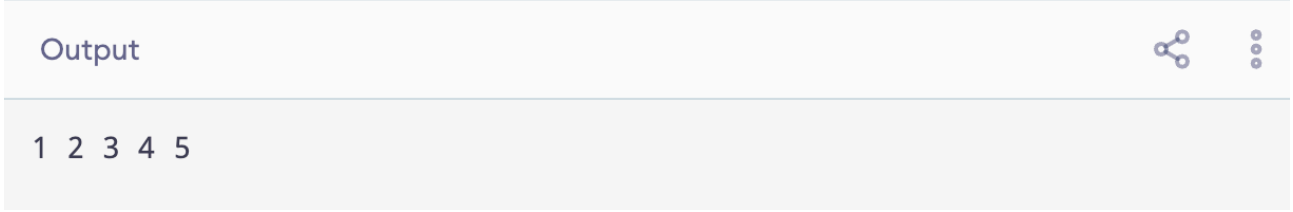
    for(i = 0; i < n - 1; i++)
    {
        min = i;

        for(j = i + 1; j < n; j++)
        {
            if(a[j] < a[min])
                min = j;
        }

        swap(&a[min], &a[i]);
    }
}

void swap(int *x, int *y)
{
    int t = *x;
    *x = *y;
    *y = t;
}
```

Output:



Code:

```
#include <stdio.h>

int main()
{
    int arr[5] = {44, 33, 11, 22, 55};
    int n = 5;
    int i, j, temp;

    for(i = 0; i < n - 1; i++)
    {
        for(j = 0; j < n - i - 1; j++)
        {
            if(arr[j] > arr[j + 1])
            {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }

    printf("Array after implementing Bubble Sort: ");

    for(i = 0; i < n; i++)
    {
        printf("%d  ", arr[i]);
    }

    return 0;
}
```

Output:



Code:

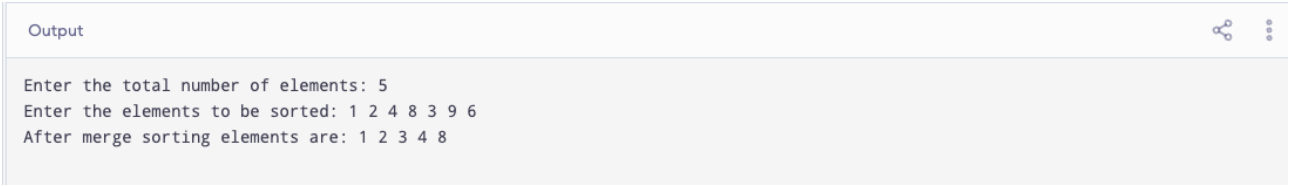
```
#include <iostream>
using namespace std;
#define MAX 50
void merge(int arr[], int low, int mid, int high);
void mergeSort(int arr[], int low, int high);
int main()
{
   int arr[MAX], i, n;
   cout << "Enter the total number of elements: ";
   cin >> n;
   cout << "Enter the elements to be sorted: ";
   for(i = 0; i < n; i++)
   {
      cin >> arr[i];
   }
   mergeSort(arr, 0, n - 1);
   cout << "After merge sorting elements are: ";
   for(i = 0; i < n; i++)
   {
      cout << arr[i] << " ";
   }
   return 0;
}
void mergeSort(int arr[], int low, int high)
{
   int mid;
   if(low < high)
   {
      mid = (low + high) / 2;
      mergeSort(arr, low, mid);
      mergeSort(arr, mid + 1, high);
      merge(arr, low, mid, high);
   }
}
void merge(int arr[], int low, int mid, int high)
{
   int i, m, k, l;
   int temp[MAX];
   l = low;
   i = low;
   m = mid + 1;
   while(l <= mid && m <= high)
   {
      if(arr[l] <= arr[m])
      {
         temp[i] = arr[l];
         l++;
      }
      else
      {
         temp[i] = arr[m];
         m++;
      }

      i++;
   }
  if(l > mid)
   {
```

```
        for(k = m; k <= high; k++)
        {
            temp[i] = arr[k];
            i++;
        }
    }
    else
    {
        for(k = l; k <= mid; k++)
        {
            temp[i] = arr[k];
            i++;
        }
    }

    for(k = low; k <= high; k++)
    {
        arr[k] = temp[k];
    }
}
```

Output:



Code:

```
#include <iostream>
using namespace std;

int main()
{
    int a[2][2], b[2][2], c[2][2];
    int i, j;

    int m1, m2, m3, m4, m5, m6, m7;

    cout << "Enter the 4 elements of first matrix: ";

    for(i = 0; i < 2; i++)
    {
        for(j = 0; j < 2; j++)
        {
            cin >> a[i][j];
        }
    }

    cout << "Enter the 4 elements of second matrix: ";

    for(i = 0; i < 2; i++)
    {
        for(j = 0; j < 2; j++)
        {
            cin >> b[i][j];
        }
    }

    cout << "\nThe first matrix is:\n";

    for(i = 0; i < 2; i++)
    {
        for(j = 0; j < 2; j++)
        {
            cout << a[i][j] << "\t";
        }
        cout << endl;
    }

    cout << "\nThe second matrix is:\n";

    for(i = 0; i < 2; i++)
    {
        for(j = 0; j < 2; j++)
        {
            cout << b[i][j] << "\t";
        }
        cout << endl;
    }

    // Strassen's seven multiplications

    m1 = (a[0][0] + a[1][1]) * (b[0][0] + b[1][1]);

    m2 = (a[1][0] + a[1][1]) * b[0][0];

    m3 = a[0][0] * (b[0][1] - b[1][1]);
```

```
    m4 = a[1][1] * (b[1][0] - b[0][0]);

    m5 = (a[0][0] + a[0][1]) * b[1][1];

    m6 = (a[1][0] - a[0][0]) * (b[0][0] + b[0][1]);

    m7 = (a[0][1] - a[1][1]) * (b[1][0] + b[1][1]);

    // Calculate result matrix

    c[0][0] = m1 + m4 - m5 + m7;

    c[0][1] = m3 + m5;

    c[1][0] = m2 + m4;

    c[1][1] = m1 - m2 + m3 + m6;

    cout << "\nAfter multiplication using Strassen's Algorithm:\n";

    for(i = 0; i < 2; i++)
    {
        for(j = 0; j < 2; j++)
        {
            cout << c[i][j] << "\t";
        }
        cout << endl;
    }

    return 0;
}
```

Output:

